higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

INTRODUCTION TO SYSTEMS DEVELOPMENT
(Second Paper**)**
NQF LEVEL 2

(10041022)

28 November 2019 (Y-Paper)
13:00–17:00

This question paper consists of 18 pages.

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TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

INSTRUCTIONS TO STUDENTS AND LECTURERS

1. Students MUST be seated in the computer laboratory 30 minutes before the start of the examination to ensure that all computers have the appropriate programs.
2. These questions must be completed on a computer by making use of MS Office 2003 or a later version and VB.NET 2005 or VB.NET Express or VB.NET 2008 or later.
3. Students must make sure that they print their work immediately after they completed a question. Ensure that your examination number is on ALL printouts. NO QUESTIONS WITHOUT A PRINTED EXAMINATION NUMBER WILL BE MARKED.
4. In the event of problems such as: a power failure; computer breakdown or a printer breakdown, the invigilator will make the necessary arrangements to continue with the examination process once the problem has been resolved. Students will not be penalised with time lost under these circumstances.
5. All work on the computer should be saved at regular intervals to prevent loss of work. NO additional time may be allowed for work lost due to lack of saving.

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6. No student may print his/her work for another student, or make a memory stick available to another student or access another student's work on the network. Any attempt to access any information from or transfer information to another student in whatever manner is a contravention of the examination rules and regulations and will be viewed in an extremely serious light.

7. All PRINTOUTS and screen prints done during the session MUST be handed in.

8. Make sure that your EXAMINATION NUMBER appears on all printouts.

9. Students may NOT copy the source code from VB.NET to MS Word and then print it from the word processor.

10. Steps to print a VB.NET form or any other screen:

    10.1 Run the solution.

    10.2 Hold the ALT key down and press the print screen button (This will copy the active screen).

    10.3 Open MS Word and paste the form/screen in a new document.

    10.4 Add your EXAMINATION NUMBER to the top right-hand side of your page.

    10.5 Print the page.

11. You are allowed to make use of the help files in VB.NET.

12. The following files will appear on your hard drive in the folders:

    12.1 QUESTION 1
    … \CarServiceBooking

    IMAGE: car.jpg

    … \IMAGES

    12.2 QUESTION 2
    … \BudgetCalculator

    12.3 QUESTION 3
    … \ShopCardDiscount

    12.4 QUESTION 4
    … \WCTeamComparison

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QUESTION 1

The following questions refer to the VB.NET program called ***CarServiceBooking*** that has been saved on your computer.

A car servicing company asks for your services as a User Interface expert to create a user interface for their new software which will allow their employees to make service bookings easier.

The employee enters the model and year of the car to be serviced, chooses the service type, and then selects the date of the last service and the new service booking date. When the employee is happy with his/her entries and selections, he/she clicks a 'Confirm Booking' button which displays a confirmation message. The employee can also click on a button to clear all entries and selections made.

NB: NO CODE IS REQUIRED FOR THIS QUESTION, YOU ARE ONLY REQUIRED TO CREATE THE GRAPHICAL USER INTERFACE

Use FIGURE 1 below and the instructions which follow as a guide to help you complete the interface.

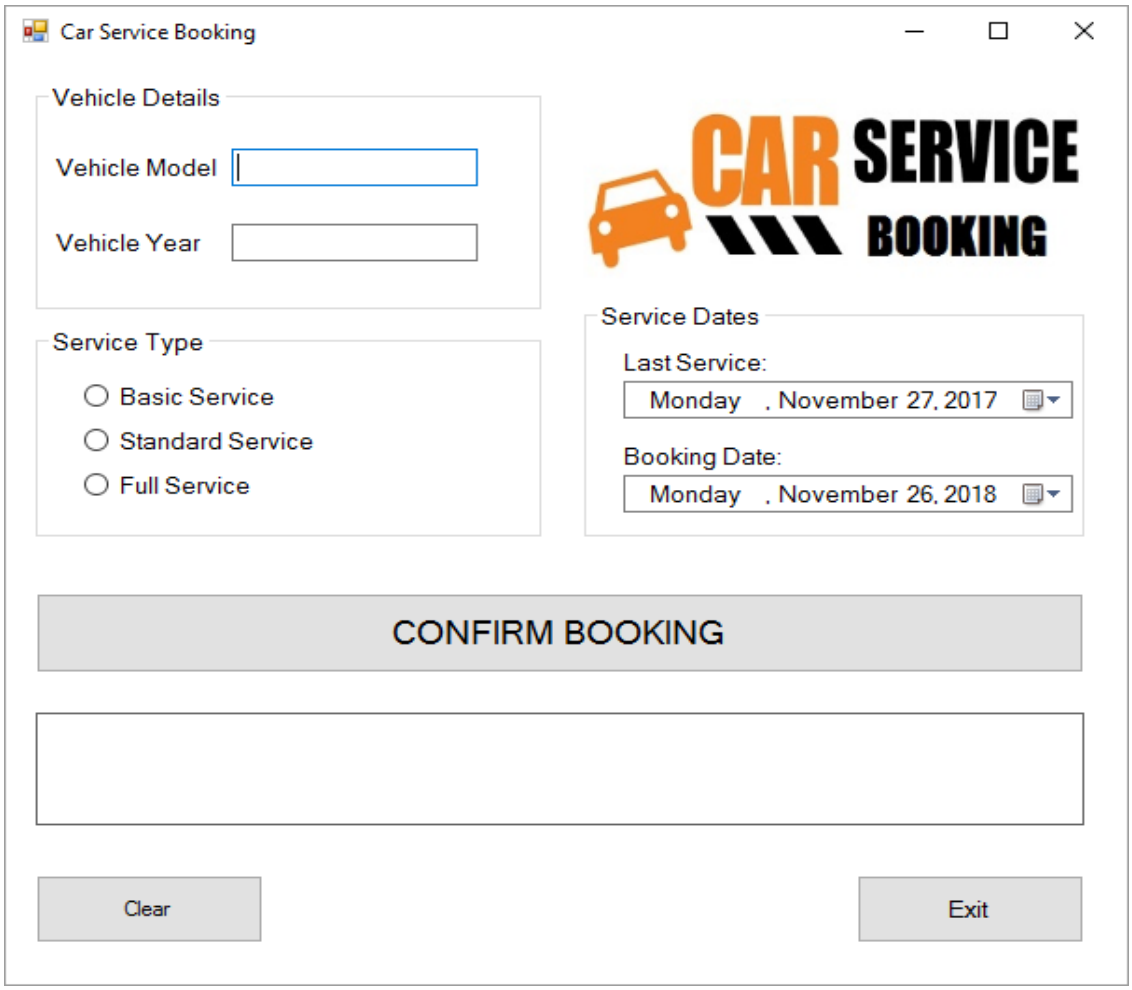


FIGURE 1

A blank interface has been saved on your computer. Open the program ***Car Service Booking*** and follow the instructions below to complete the interface.

1.1 Add a PICTURE BOX to the form as part of the heading:

CONTROL	NAME	IMAGE
PictureBox	picCar	Use the image saved on your computer as 'car.jpg'

(2)

1.2 Add THREE GROUP BOXES to the form, to group the Vehicle Details, Service Types and Service Dates:

CONTROL	NAME	TEXT
Group Box	grpVehicleDetails	Vehicle Details
Group Box	grpServiceType	Service Type
Group Box	grpServiceDates	Service Dates

(3)

1.3 Add FIVE LABELS to the form, two for the car details, two for the service dates, and one to display a confirmation message:

CONTROL	NAME	TEXT	PROPERTIES
Label	lblVehicleMod	Vehicle Model	Default
Label	lblVehicleYear	Vehicle Year	Default
Label	lblLastServ	Last Service	Default
Label	lblBookDate	Booking Date	Default
Label	lblDisplay	*(Blank)*	**Border:** Fixed Single **Size:** See Figure 1

(7)

1.4 Add TWO TEXT BOXES to the form, to enter the car model and car year:

CONTROL	NAME	TEXT
Text Box	txtVehicleModel	(Blank)
Text Box	txtVehicleYear	(Blank)

(2)

1.5 Add THREE RADIO BUTTONS to the form, to indicate the service type:

CONTROL	NAME	TEXT
Radio Button	radBasic	Basic
Radio Button	radStandard	Standard
Radio Button	radFull	Full

(3)

1.6 Add TWO DATETIMEPICKERS to the form, to select the last service date and the new service date:

CONTROL	NAME
DateTimePicker	dtpLastService
DateTimePicker	dtpNewSerive

(4)

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1.7 Add THREE BUTTONS to the form, one to make the booking, one to clear all the fields on the form, and one to exit the program:

CONTROL	NAME	TEXT	Font Size
Button	btnBooking	Confirm Booking	Default
Button	btnClear	Clear	Default
Button	btnExit	Exit	Default

(3)

1.8 Change the text in the title of the form to your EXAMINATION NUMBER and make a PRINTOUT of the form. (2)

[26]

QUESTION 2

The following questions refer to the VB.NET program called ***BudgetCalculator*** that has been saved on your computer.

You have recently been going through financial difficulty, and as you are a programmer, you decide to create a program to help you keep track of your finances. Your program has two main sections, namely, 'Income' and 'Expenses'. The income section consists of the following fields:

- Salary
- Other Income

Your program then uses these fields to calculate your *total monthly income*.

The expenses section consists of the following fields:

- Rent
- Groceries
- Water and Electricity
- Entertainment

Your program then uses these fields to calculate your *total monthly expenses*.

The program then calculates your *net income* using these two values.

Use the IPO chart below to help you with the program:

INPUT	PROCESSING	OUTPUT
Salary OtherIncome Rent Groceries WaterAndElectricity Entertainment	**TotalIncome** = Salary + Other Income **TotalExpenses**= Rent + Groceries + WaterAndElectricity + Entertainment **NetIncome** = TotalIncome - TotalExpenses	TotalIncome TotalExpenses NetIncome

FIGURE 2 below is a screenshot of what the user interface looks like. The completed interface has been saved on your computer. Open the program ***BudgetCalculator*** and follow the instructions to complete the program:

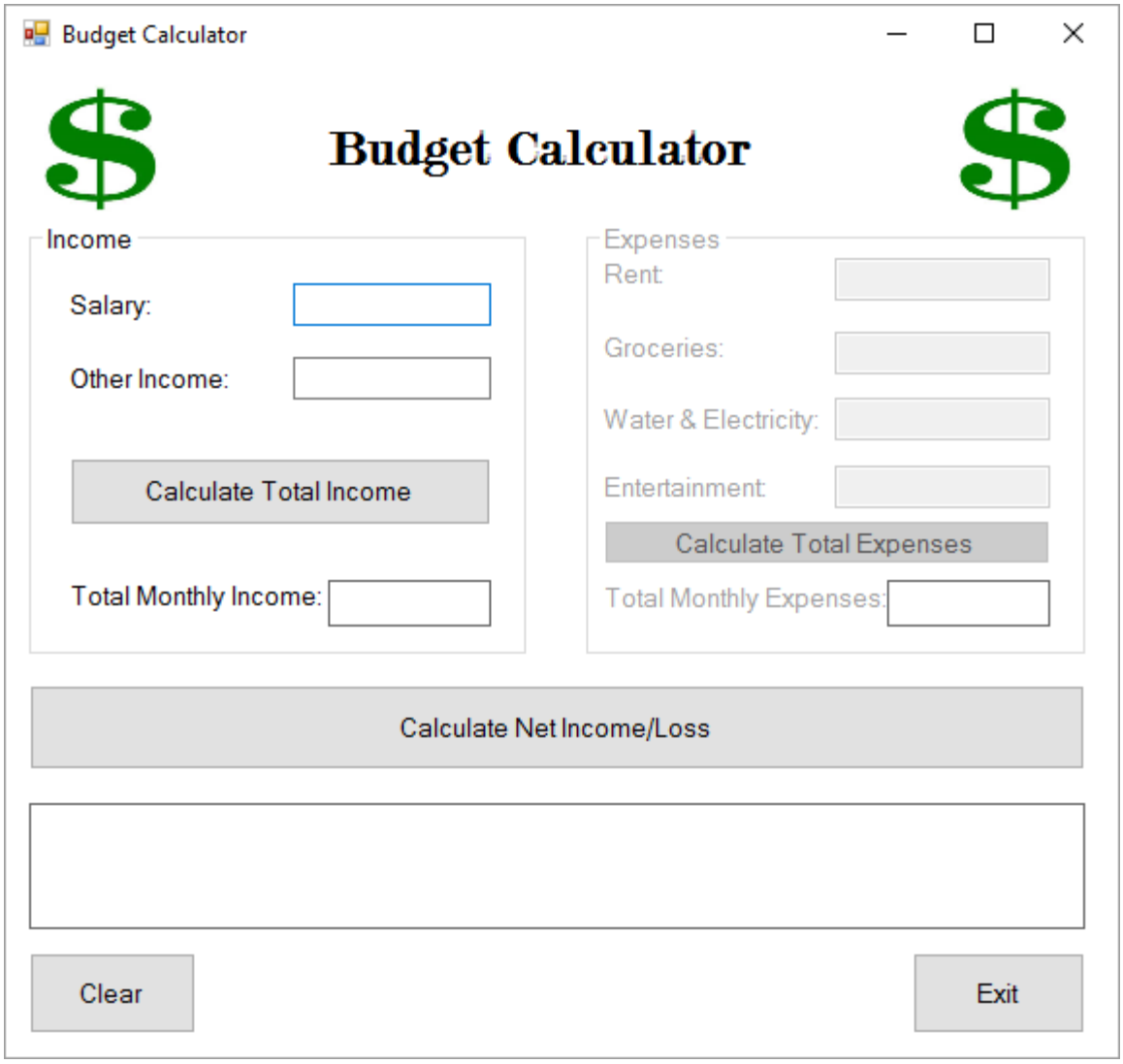
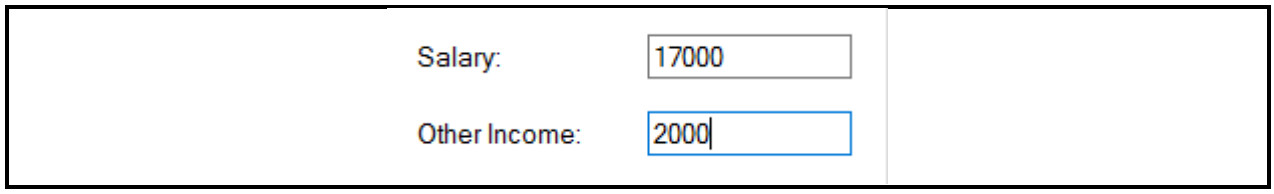


FIGURE 2

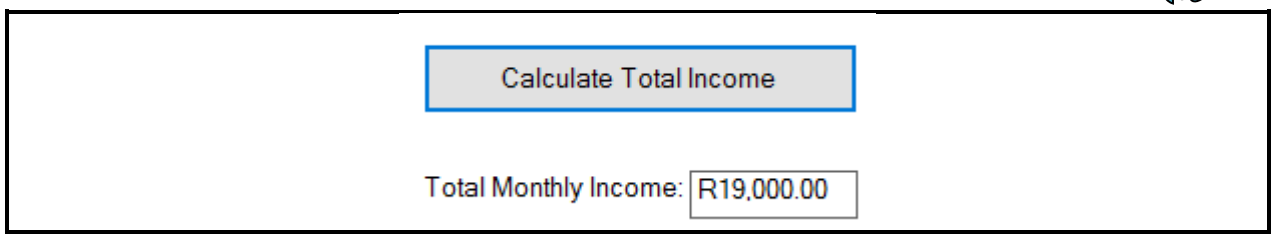
The following steps explain how to use the program in order to calculate your net income:

STEP 1 RUN the program and enter your *Salary* and *Other Income*:

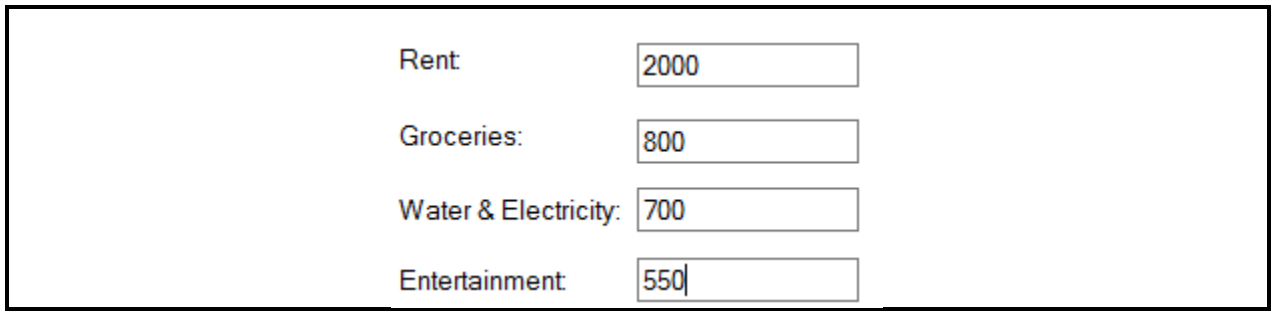


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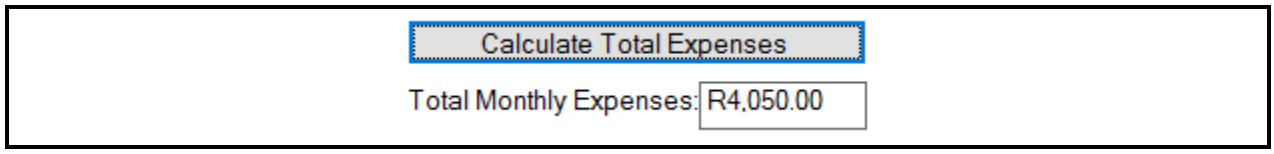
STEP 2 Click the *Calculate Total Income* button to calculate and display the *Monthly Income* as follows (This also enables the *Expenses* group box):



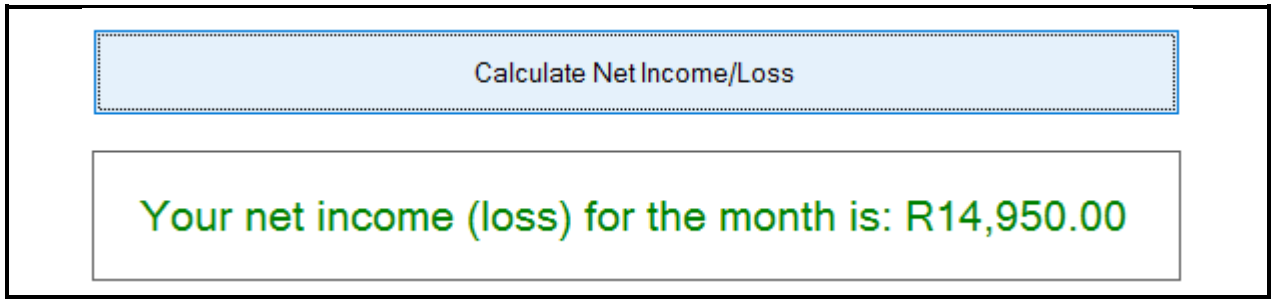
STEP 3 Enter your *Rent, Groceries, Water and Electricity* and *Entertainment*:



STEP 4 Click the *Calculate Total Expenses* button to calculate and display the *Total Expenses* as follows:



STEP 5 Click the *Calculate Net Income* button to calculate and display your *Net Income* as follows:



2.1 Declare the *total income* and *total expenses* as *global* variables. (2)

2.2 Add the following program code to the click event of the *Calculate Total Income* button:

2.2.1 Declare ALL the required variables. (2)

2.2.2 Convert and assign the values from the salary and other income text boxes to the appropriate variables. (2)

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2.2.3 Calculate and display the *total income* in the appropriate label. *See Step 2 above.* (4)

2.2.4 Enable the *Expenses* group box. (1)

2.3 Add the following program code to the click event of the *Calculate Total Expenses* button:

2.3.1 Declare ALL the required variables. (2)

2.3.2 Convert and assign the values from the *rent, groceries, water and electricity* and *entertainment* text boxes to the appropriate variables. (4)

2.3.3 Calculate and display the *total expenses* in the appropriate label. *See Step 4 above.* (4)

2.4 Add the following program code to the click event of the *Calculate Net Income* button:

2.4.1 Declare ALL the required variables. (1)

2.4.2 Calculate and display the *net income* in the appropriate label as shown in Step 5 above. (4)

2.5 Clicking the *Clear Button* should clear ALL the text boxes and the display labels, and also disable the *Expenses* groupbox. (3)

2.6 Clicking the *Exit Button* should close the program. (1)

Make a PRINTOUT of the source code.

[30]

QUESTION 3

The following questions refer to the VB.NET program called ***ShopCardDiscount*** that has been saved on your computer.

A small clothing store needs to help their customers calculate the discount they could get if they had a shop card and were members of different loyalty programs. The customers can have one of three types of shop cards as follows:

- Bronze Card (5% discount)
- Silver Card (7% discount)
- Gold Card (12% discount)

Their customers can also choose to join their loyalty programs, which are:

- Green Loyalty Member (2% discount on amount more than R2000)
- Prime Member (15% discount on amounts more than R10 000)

NOTE: A customer can be both a Green Loyalty Member AND a Prime Member – i.e. A customer that is both would get a combined discount of 17%

They ask you to design a program which will calculate the *total discount* the customer would get based on a *total amount* spent.

Use the IPO chart below to help you with the program:

INPUT	PROCESSING	OUTPUT
Total Spent Select Card Type Select Loyalty Benefits	**Card Discounts:** • Bronze – 5% • Silver – 7% • Gold – 12% **Loyalty Discounts:** • Green Member – 2% • Prime Member – 15% **TotalDiscountPercnt** = CardDiscount + LoyaltyDiscount **TotalDiscount** = TotalSpent * TotalDiscountPercent	TotalDiscount

FIGURE 3 below is a screenshot of what the user interface looks like. The completed interface has been saved on your computer. Open the solution ***ShopCardDiscount*** and follow the instructions to complete the program:

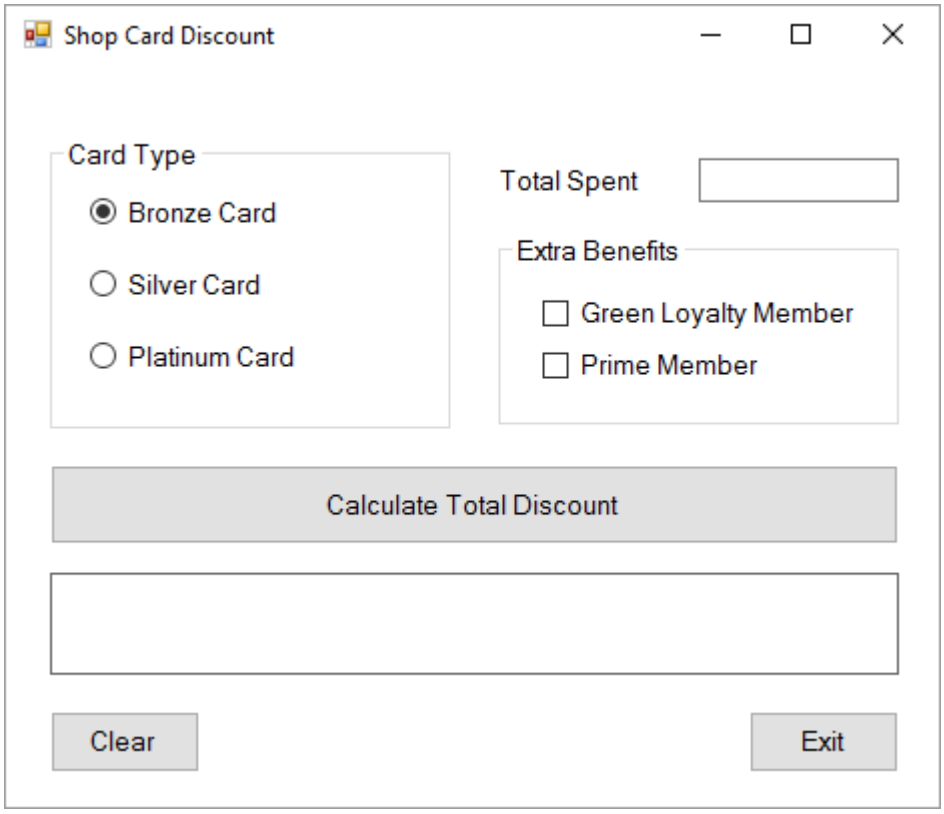


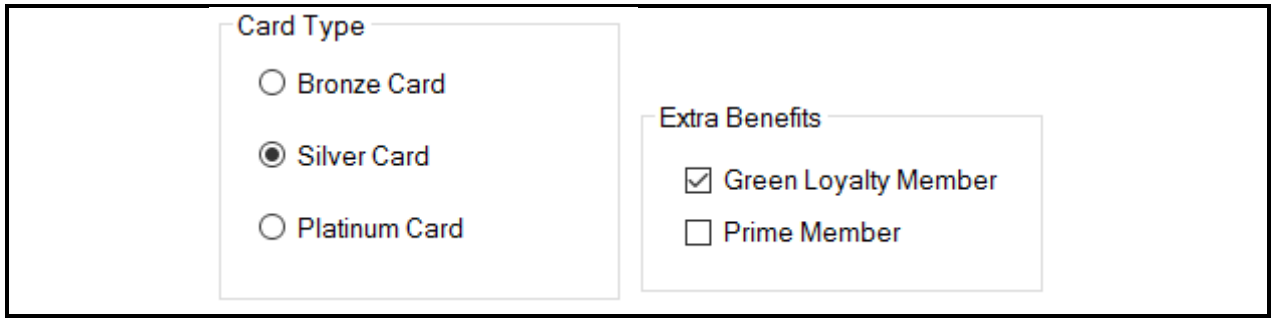
FIGURE 3

The following steps explain how to use the program in order to calculate the total discount:

STEP 1 RUN the program and enter the *total amount* spent:

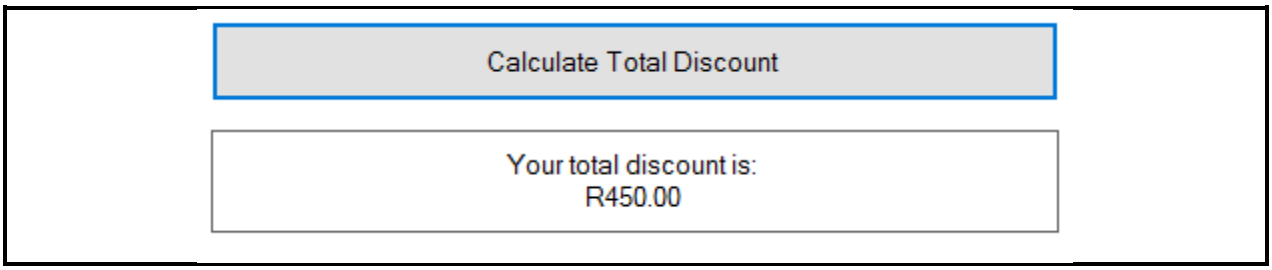


STEP 2 Select the *card type* and *loyalty benefits:*



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STEP 3 Click the *Calculate Total Discount* button to calculate and display the *Total Discount amount* as follows:



3.1 Add the following program code to the click event of the *Calculate Total Discount* button:

3.1.1 Declare ALL the required variables. (3)

3.1.2 Convert and assign the value from the *Amount Spent* text box to the appropriate variable. (1)

3.1.3 Determine the *card discount percentage* based on the radio button selection. (4)

3.1.4 Determine the *loyalty discount percentage* based on the check box selection(s). (4)

3.1.5 Calculate the *Total Discount Percentage.* (2)

3.1.6 Calculate the *total discount amount* based on the *amount spent* and *total discount percentage.* (3)

3.1.7 Display the *total discount* in the appropriate label as shown in Step 3 above. (3)

3.2 Clicking the *Clear Button* should clear the text box, radio buttons, check boxes and the display label. (4)

3.3 Clicking the *Exit Button* should close the program. (1)

Make a PRINTOUT of the source code.

[25]

QUESTION 4

The following questions refer to the VB.NET program called ***WCTeamComparison*** that has been saved on your computer.

You are an avid soccer fan, and since you're a programmer as well, you decide to put your programming skills to work to help you compare teams playing in the soccer world cup.

The program should allow you to choose between comparing either two, or three teams with each other. The user enters the details for each team (Games played, Goals Forward, Goals Against and Number of Wins) and then clicks a *Compare* button. The programs then uses the information entered to calculate the win percentage and goal difference of each team and then displays all the information for each team in a list box.

The program then also decides which team is the best based on the win percentage, and also displays this information in the list box.

FIGURE 4 on the next page is a screenshot of what the user interface looks like:

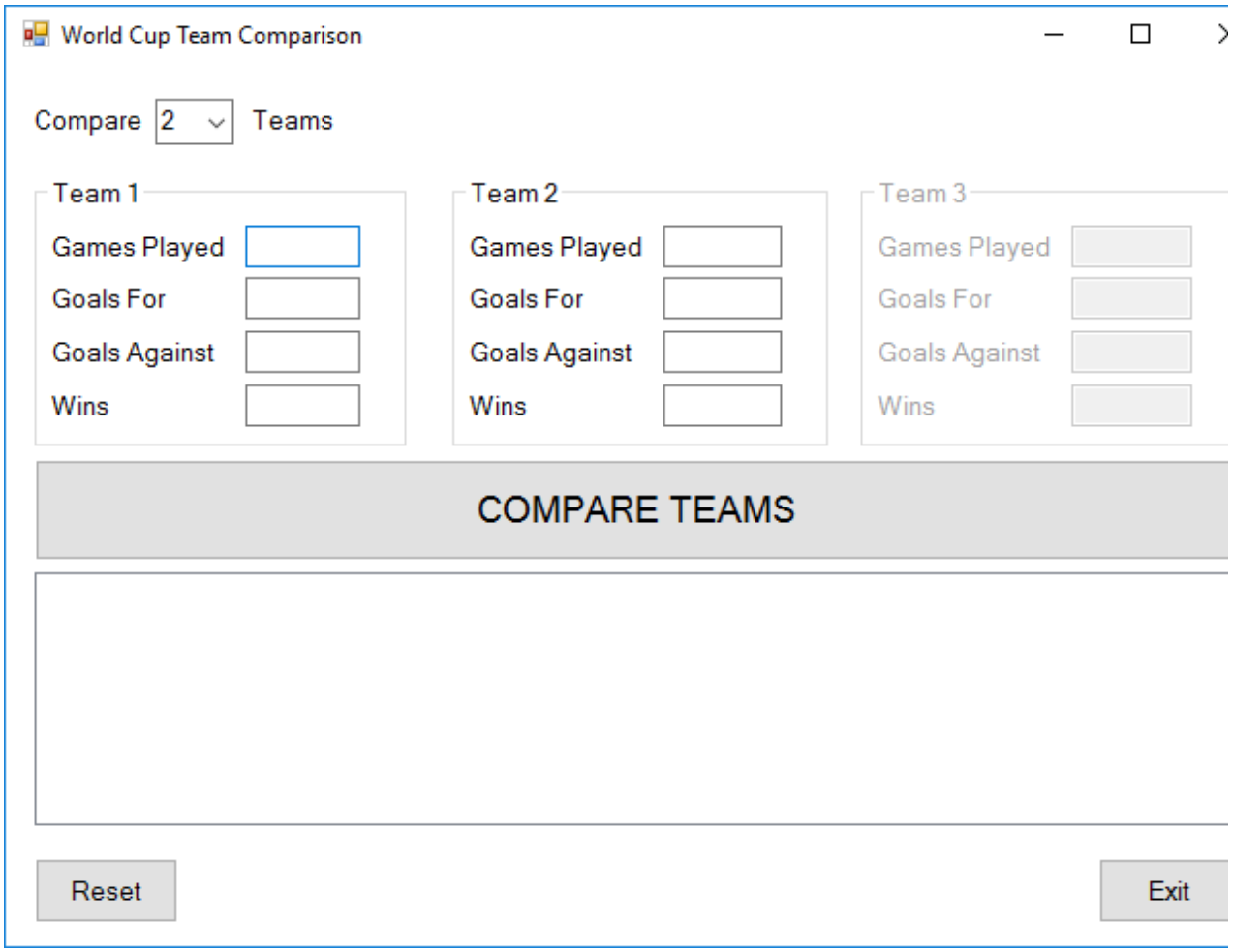


FIGURE 4

WORKING OF THE PROGRAM EXPLAINED:

The following steps explain how to use the program:

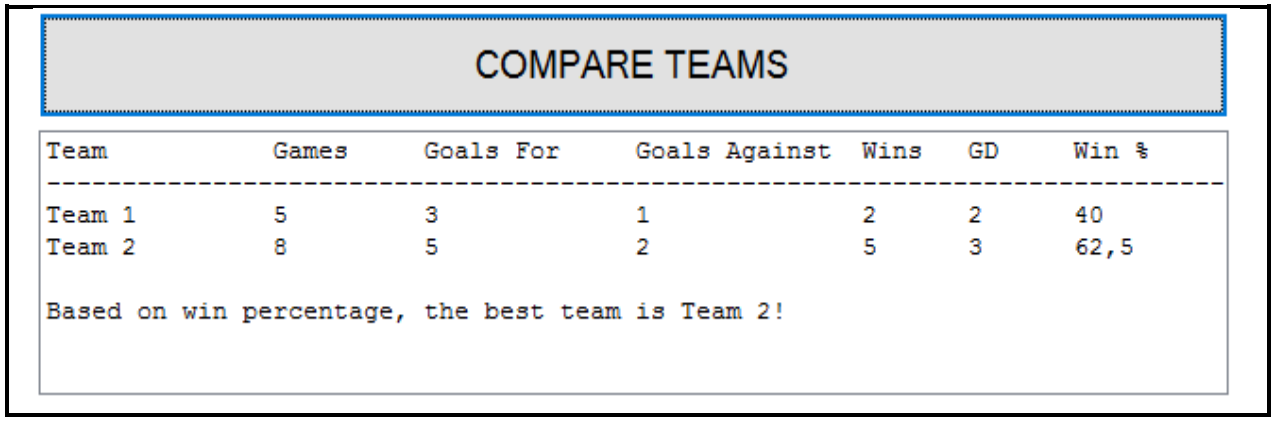
STEP 1 RUN the program and select how many teams to compare:



STEP 2 Enter the required information for the teams:



STEP 3 Click the *Compare Teams* button to display the result:



STEP 4 Clicking the *Reset* button resets all the textboxes, clears the list box, selects the first item in the combo box, and moves the cursor to the *Team 1 Games Played* text box.

STEP 5 Clicking the *Exit* button closes the program.

You have already designed and started coding the application, but it contains numerous syntax and logical errors, and you now have to debug the source code.

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Below is a PRINTOUT of the program code with all the syntax and logical errors still present:

```
…
Public Class Form1
    'Examination Number Here
    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
                        System.EventArgs) Handles MyBase.Load
        'add items to combobox
        cboTeams.Items.Input(2)
        cboTeams.Items.Add(3)
        'select first item
        cboTeams.SelectedIndex = 0
    End Sub

Private Sub btnCompare_Click(ByVal sender As System.Object, ByVal e
As System.EventArgs) Handles btnCompare.Click
        Dim intGamesT1 As Integer = 0
        Dim intGoalsForT1 As Integer = 0
        Dim intGoalsAgainstT1 As Integer = 0
        Dim intWinsT1 As Integer = 0
        Dim intGoalDifferenceT1 As Integer = 0
        Dim decWinPercentT1 As Decimal = 0

        Dim intGamesT2 As Integer = 0
        Dim intGoalsForT2 As Integer = 0
        Dim intGoalsAgainstT2 As Integer = 0
        Dim intWinsT2 As Integer = 0
        Dim intGoalDifferenceT2 As Integer = 0
        Dim decWinPercentT2 As Decimal = 0

        Dim strWinner As String = ''
        Dim decBestPercent As Decimal = 0

        'assign textbox values to variables
        intGamesT1 = CInt(txtGamesT1)
        intGoalsForT1 = txtGoalsForT1.Text
        intGoalsAgainstT1 = CInt(txtGoalsAgainstT1.Text)
        intWinsT1 = CInt(txtWinsT1.Text)

        intGamesT2 = CInt(txtGamesT2.Text)
        intGoalsForT2 = CInt(txtGoalsForT2.Text)
        intGoalsAgainstT2 = CInt(txtGoalsAgainstT2.Text)
        intWinsT2 = CInt(txtWinsT2.Text)
        calculate goal difference
        intGoalDifferenceT1 = intGoalsForT1 + intGoalsAgainstT1
        intGoalDifferenceT2 = intGoalsForT2 - intGoalsAgainstT2
        'calculate win percentage
        decWinPercentT1 = CDec((intWinsT1 / intGamesT1) * 100)
        decWinPercentT2 = CDec((intWinsT2 / intGamesT2) * 100)
        'determine winner
        If decWinPercentT1 < decWinPercentT2 Then
            strWinner = 'Team 1'
            decBestPercent = decWinPercentT1

        Else
            strWinner = 'Team 2'
            decBestPercent = decWinPercentT2
```

```
        End If
        'display results
        lstDisplay.Items.Add('Team'.PadRight(15)
'Games'.PadRight(10) & 'Goals For'.PadRight(14) & 'Goals
Against'.PadRight(15) & 'Wins'.PadRight(7) & 'GD'.PadRight(7) & 'Win
%')
        lstDisplay.Items.Add('----------------------------------------
----------------------------------------')
        lstDisplay.Items.Add('Team 1'.PadRight(15) &
intGamesT1.ToString.PadRight(10) &
intGoalsForT1.ToString.PadRight(14) &
intGoalsAgainstT1.ToString.PadRight(15) &
intWinsT1.ToString.PadRight(7) &
intGoalDifferenceT1.ToString.PadRight(7) & decWinPercentT1.ToString)
        lstDisplay.Items.Add('Team 2'.PadRight(15) &
intGamesT2.ToString.PadRight(10) &
intGoalsForT2.ToString.PadRight(14) &
intGoalsAgainstT2.ToString.PadRight(15) &
intWinsT2.ToString.PadRight(7) &
intGoalDifferenceT2.ToString.PadRight(7) & decWinPercentT2.ToString)
        'only add in Team 3 if required
         If grpTeam3.Enabled = False Then
            Dim intGamesT3 As Integer = 0
            Dim intGoalsForT3 As Integer = 0
            Dim intGoalsAgainstT3 As Integer = 0
            Dim intWinsT3 As Integer = 0
            Dim intGoalDifferenceT3 As Integer = 0
            Dim decWinPercentT3 As Decimal = 0

            intGamesT3 = CInt(txtGamesT3.Text)
            intGoalsForT3 = CInt(txtGoalsForT2.Text)
            intGoalsAgainstT3 = CInt(txtGoalsAgainstT3.Text)
            intWinsT3 = CInt(txtWinsT3.Text)
            intGoalDifferenceT3 = intGoalsForT3 - intGoalsAgainstT3
            decWinPercentT3 = CDec((intWinsT3 / intGamesT3) / 100)
            If decWinPercentT3 > decBestPercent Then
                strWinner = 'Team 3'
            End If
            lstDisplay.Items.Add('Team 3'.PadRight(15) &
intGamesT3.ToString.PadRight(10) &
intGoalsForT3.ToString.PadRight(14) &
intGoalsAgainstT3.ToString.PadRight(15) &
intWinsT3.ToString.PadRight(7) &
intGoalDifferenceT3.ToString.PadRight(7) & decWinPercentT3.ToString)

        End If
        lstDisplay.Items.Add('')
        lstDisplay.Text('Based on win percentage, the best team is '
& strWinner & '!')

    End Sub

Private Sub btnNext_Click(sender As Object, e As EventArgs) Handles
btnNext.Click
        'Clear form controls
        radChicken.Checked = False
        radBeef.Checked = False
        radVeggie.Checked = False

        chkCheese.Checked = False
        chkSauce.Checked = False
        chkChillies.Checked = False
```

```
        cboDiscount.SelectedIndex = 0
        txtAmountPaid.Empty()

        lblTotalDue = ''
        lblChange.Text = ''
    End Sub

    Private Sub cboTeams_SelectedIndexChanged(ByVal sender As
System.Object, ByVal e As System.EventArgs) Handles
cboTeams.SelectedIndexChanged
        'disable Team 3 group box based on combobox selection
        If cboTeams.SelectedIndex = 0 Then
            grpTeam3.Enabled = False
        ElseIf cboTeams.SelectedIndex = 1 Then
            grpTeam3.Enabled = True

    End Sub

Private Sub btnReset_Click(ByVal sender As System.Object, ByVal e As
System. EventArgs) Handles btnReset.Click
        'reset controls
        txtGamesT1.Empty()
        txtGamesT2.Clear()
        txtGamesT3.Clear()
        txtGoalsAgainstT1.Clear()
        txtGoalsAgainstT2.Clear()
        txtGoalsAgainstT3.Clear()
        txtGoalsForT1.Clear()
        txtGoalsForT2.Clear()
        txtGoalsForT3.Clear()
        txtWinsT1.Clear()
        txtWinsT2.Clear()
        txtWinsT3.Clear()

        lstDisplay.Items.Clear()
        cboTeams.SelectedIndex = 0
        txtGamesT1.Focus()
    End Sub

Private Sub btnExit_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnExit.Click
        'exit program
        Close()
End Sub.
```

4.1 The completed program has been saved on your computer, but the program is not functioning properly.

Open the program ***WCTeamComparison*** and correct all the syntax AND logical errors. (16)

4.2 Add your EXAMINATION NUMBER to the source code as indicated, and make a PRINTOUT of the source code. (1)

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4.3 Add you Examination Number to the title bar of the form, and then use the program to determine which of the following TWO teams are better:



	Team 1	Team 2
Games played	15	10
Goals For	21	15
Goals Against	18	10
Wins	10	6

Make a PRINTOUT of the form showing the correct result. (2)
[19]

TOTAL: 100

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